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## **HSE Digital Solutions is the Path for Injury Free Operations – Digital PPE**

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### **Abstract**

Traditional HSE tools have plateaued in effectiveness, with minimal improvements in incident rates. This paper presents a practical deployment of AI-driven digital safety tools—including real-time video analytics for red zone detection, the Platform for Automated Register Intractability (PARI), and auto-translated multilingual observation cards—to enhance safety and operational efficiency at SLB drilling sites. The integrated system applies artificial intelligence and automation to reduce human error, eliminate language barriers, and streamline compliance tracking. Video analytics utilize rig floor data and camera feeds to detect red zone violations and trigger real-time audio-visual alerts. Platform for Automated Register Intractability (PARI) automates register creation by extracting data from certificates, while digital observation cards enable multilingual, mobile safety reporting with automatic translation and seamless integration into daily reports. All solutions operate on cloud-based platforms, enabling real-time access, centralized data, and minimal manual intervention.

### **Introduction**

Ensuring safe operations is a fundamental requirement in oil and gas operations, particularly in high-risk activities such as drilling and lifting. While the industry has made substantial strides in health, safety, and environment (HSE) performance through structured protocols and safety training, recent global data indicate that these improvements have plateaued. Notably, post-pandemic operational trends in 2021 and 2022 have shown a resurgence in incident rates, with behavioral lapses and reduced hazard awareness cited as contributing factors. Rig-floor operations remain among the most hazardous in oil and gas drilling environments, primarily due to the dynamic interaction between heavy equipment and human presence. A persistent challenge in these environments is managing the red zone—areas around moving equipment such as the top drive, traveling block, or pipe-handling systems where personnel are at elevated risk of being struck or trapped. Despite rigorous training and procedural controls, compliance with red zone boundaries often depends on human judgment and situational awareness, both of which can be inconsistent in high-pressure operational settings.

Historically, red zone enforcement relied on visual cues and manual supervision by drillers or safety officers. However, this model introduces variability based on individual experience, fatigue, and real-time

distractions. Additionally, unintentional entry into restricted areas is difficult to detect instantly, leaving a narrow window for intervention and increasing the likelihood of high-potential incidents. Industry reports and incident investigations frequently cite poor red zone adherence as a causal or contributing factor in dropped objects, struck-by events, and other safety incidents.

As operations have grown in complexity and speed, the limitations of human-only oversight have become more evident. The need for an automated, objective system capable of monitoring, identifying, and alerting personnel to red zone breaches in real time is now recognized as critical to improving rig-floor safety.

## Objectives

As shown in Fig. 1, from the year 1985 to 2014 by International Oil & Gas Producers Association (IOGP) there is a marginal drop in the number of incidents and post that the graph is constant to some extent, which is a clear indication that the technical limits of the existing systems are near to being exhausted, hence we need new innovation like AI / Machine learning to reduce the FAR furthermore.

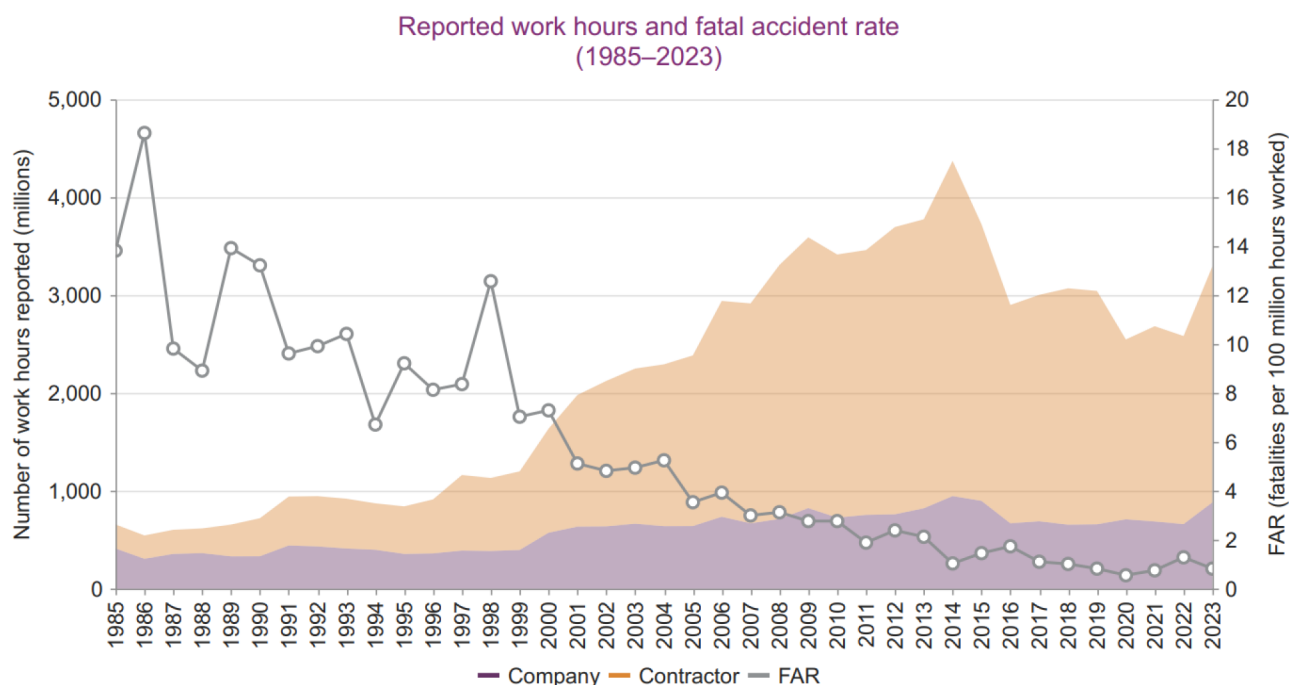


Figure 1—IOGP 2023 Reported work hours and Fatal accident rate

As shown in Fig. 2 & 3, International Oil & Gas Producers Association (IOGP) Reports stated that 26.4% of total Lost Workdays cases reported are caused because of Dropped and Caught between cases which are most likely to happen on the rig floor due to improper management of the red zones or due to Unsafe mechanical lifting practices. And 26.5% of the total reported Lost Workdays cases are in Drilling and Lifting activities in the year 2023.

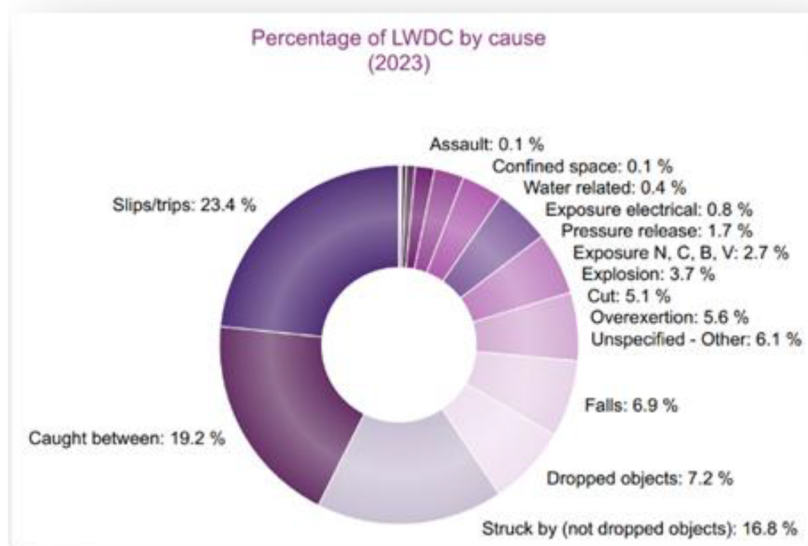


Figure 2—LWDC by cause %

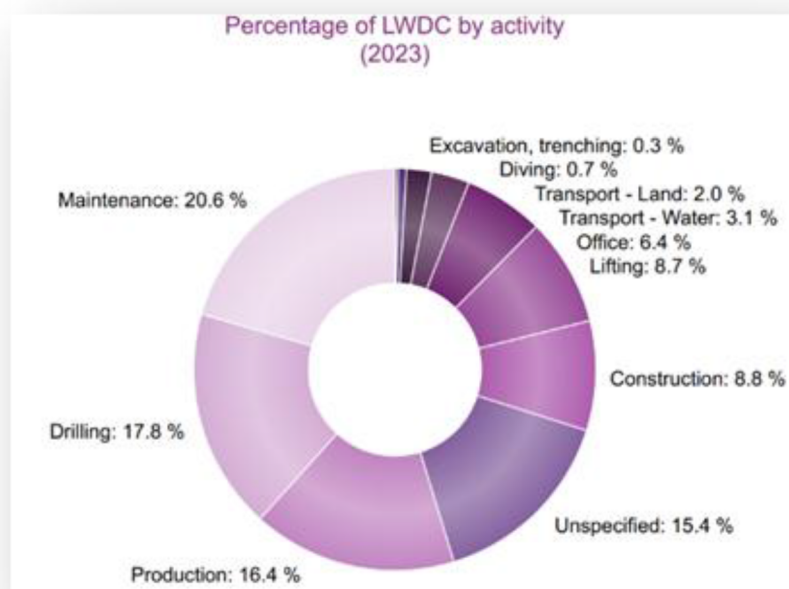


Figure 3—LWDC by activity %

As shown in Fig. 4, another report shows that out of total incident reported for onshore activities to IOGP 22.7% are during drilling activities and 3.7% of the total incidents are during lifting activities, making the total count of 26.4% of the total incidents.

As over one fourth of the incidents are happening during lifting and drilling activities in the onshore operations hence these areas safety systems need to be revamped and make them more stringent.

Hence to move ahead and put a step in the direction of new machine learned atmosphere to make the workplace a more safer place with the help of Digital PPE, these tools enable us to make the environment

safer by providing real time feedback, enabling to check the compliant equipment, breaking the borders of language across the globe and having all the data available online so that can be access and analyze at any point of the time from anywhere around the globe.

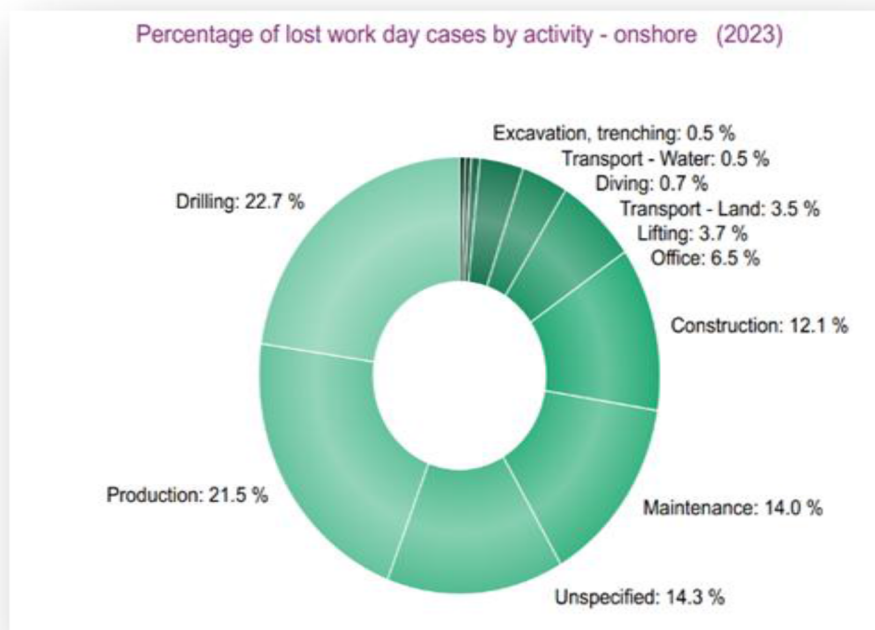


Figure 4—LWDC by activity % (Onshore)

To meet these challenges, the industry is turning to digital transformation as a means of closing existing safety performance gaps. Artificial intelligence and machine learning offer powerful capabilities for real-time monitoring, predictive analytics, and behavior-based intervention. Unlike traditional HSE tools that rely heavily on manual oversight, these technologies provide scalable solutions that standardize response and enable consistent enforcement of safety protocols across diverse operational environments.

This paper presents three integrated AI-driven digital solutions targeting key safety challenges in drilling operations: (1) video analytics for automated red zone detection, (2) a platform for automated lifting register creation and compliance tracking, and (3) a mobile, auto-translated multilingual observation's reporting system. These tools were developed based on operational risk trends to enhance situational awareness, reduce human error, and support real-time decision-making.

The first solution: The video analytics system uses AI models to monitor the rig floor and correlate personnel movement with rig activity—such as top drive speed and pressure—to detect unsafe proximity and trigger audiovisual alerts. This improves compliance consistency and promotes a proactive safety culture.

The second solution replaces manual lifting register management by extracting certificate data from uploaded PDFs, validating compliance, and storing information in a cloud-based platform. Unique serial numbers ensure equipment traceability and allow real-time readiness verification.

The third solution enables multilingual, mobile safety reporting with automatic translation, allowing all personnel to report hazards regardless of language. Reports are standardized and integrated directly into HSE dashboards, increasing participation and eliminating translation delays.



Together, these systems form a data-driven HSE framework that improves incident detection, compliance, and frontline engagement, while reducing administrative workload. The initiative also supports ESG goals by enhancing workplace safety, regulatory compliance, and operational resilience.

By delivering measurable improvements in safety performance, communication, and efficiency, the deployment of these AI-powered tools demonstrates the strategic value of digital transformation in modern drilling environments.

## Automated Video Analytics for Red Zone Detection

According to the International Association of Oil & Gas Producers (IOGP, 2023), approximately 26.4% of all reported lost workday cases resulted from dropped and caught-between incidents. These scenarios often originate from lapses in red zone enforcement and lifting operation practices on the rig floor. Furthermore, 22.7% of onshore incidents reported in the same period were linked to drilling activities. These trends underscore the critical need for enhanced safety frameworks that address not only procedural compliance but also human performance variability.

### Methodology

To mitigate the risks associated with red zone violations, a video analytics-based monitoring system has been developed and implemented. This AI-powered system integrates computer vision with real-time rig operational data to provide continuous surveillance and automated detection of unauthorized personnel movement within hazardous zones. By reducing dependency on human monitoring alone, the system enhances situational awareness and supports proactive intervention.

The system is designed to detect personnel presence using camera feeds strategically installed to cover critical zones on the rig floor. These video inputs are processed using AI models trained to recognize human figures and map their location against predefined red zone boundaries. The system simultaneously ingests rig data—such as top drive position (height), rotational speed (RPM), and operational mode (circulation or connection)—to contextualize whether the area is currently active and presents risk. When criteria for a breach are met, an audiovisual alarm is triggered immediately, warning both the driller and the person at risk.

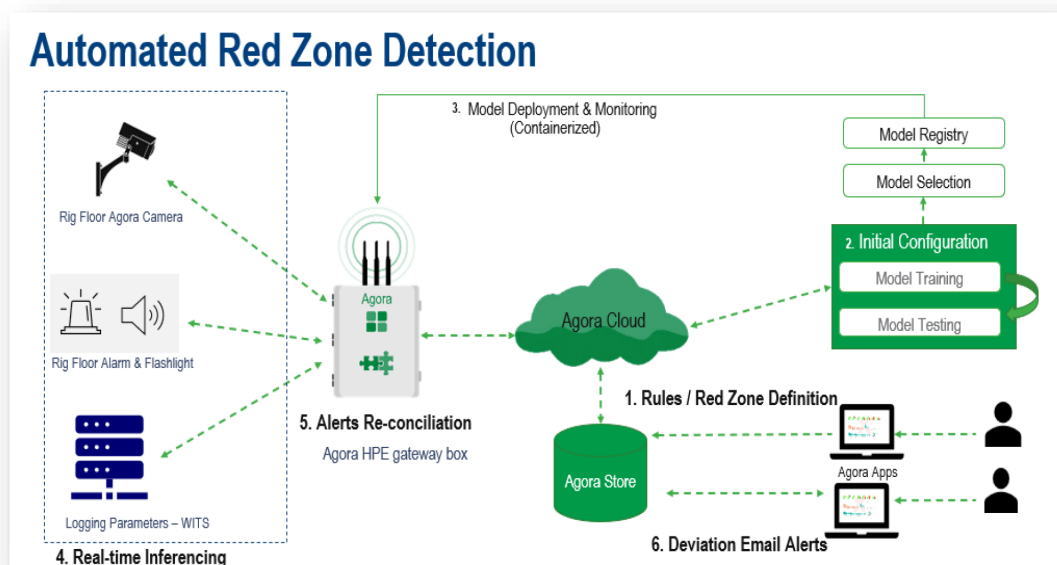


Figure 5—Automated Red Zone Detection Backend Flow

## Technical Implementation

The system architecture includes high-definition cameras connected to a local edge processing unit capable of running AI algorithms. These devices operate independently of the rig's control system but maintain data connectivity to rig sensors via programmable logic controllers (PLCs) or standard communication protocols (e.g., OPC-UA).

Detection logic is rule-based and adaptive. For example, if the top drive is below a certain height and in motion, the adjacent rig floor area is flagged as an active red zone. Personnel encroachment within this zone for more than two seconds triggers a violation alert. The system archives all events, including timestamped video and associated rig data, for post-event analysis and training use.

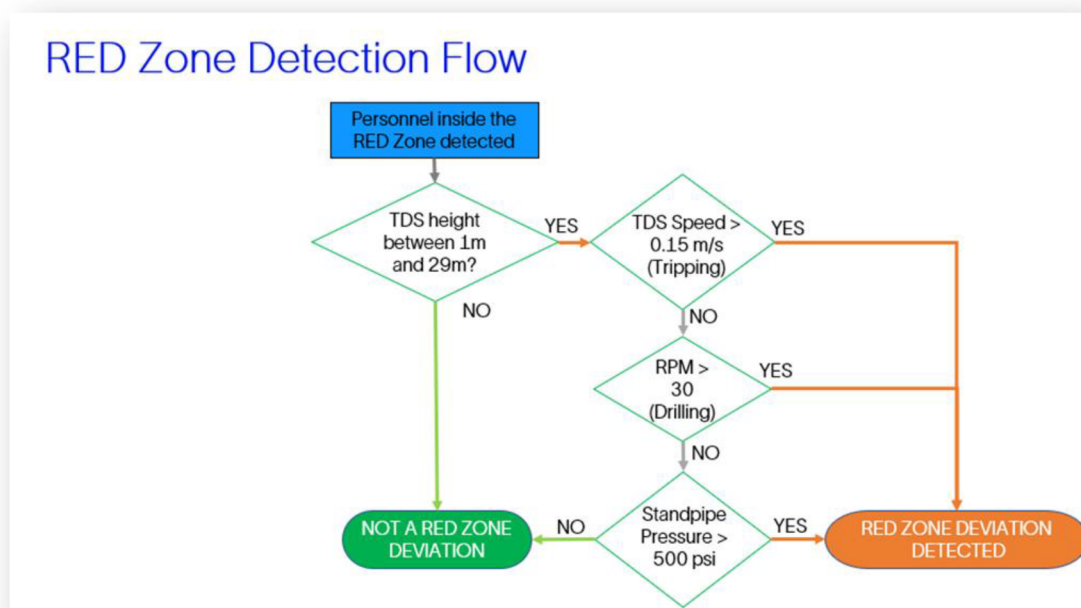


Figure 6—Red Zone Detection Flow Criteria

Data storage and analytics are managed on a cloud-based platform, which allows remote access for safety supervisors and management teams. Continuous feedback loops from field incidents are used to retrain and improve the AI models, enhancing detection accuracy over time. Additionally, the platform includes dashboards to visualize trends, heat maps of frequent violations, and compliance metrics per shift or crew.



Figure 7—Live Feeds from the Red Zone detection System

**The architecture of Video analytics is.** As the architecture of the system shown above, it will take all the data from the rig sense process it and compare it with the camera feedback and if there is any violation it

will immediately raise the alarm to alert the people and move outside the red zone at the same time it will keep the backup and the photographs and the evidence in the server so it can be viewed through websites and the notifications can be generated on weekly basis through this this artificial intelligence is going to be through the continuous learning model as well as the outcome can be customized based on the requirement and the dashboard can be created.

### **Physical Set up.**

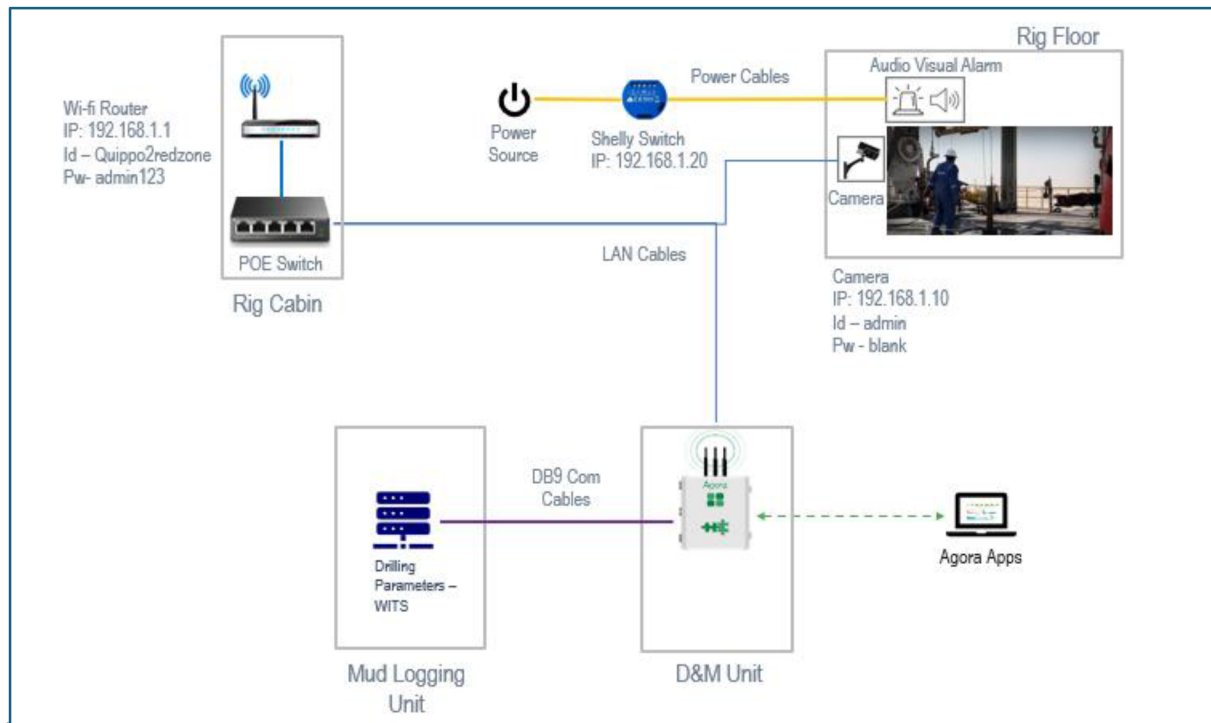


Figure 8—The connection and wiring of the camera and Agora gateway

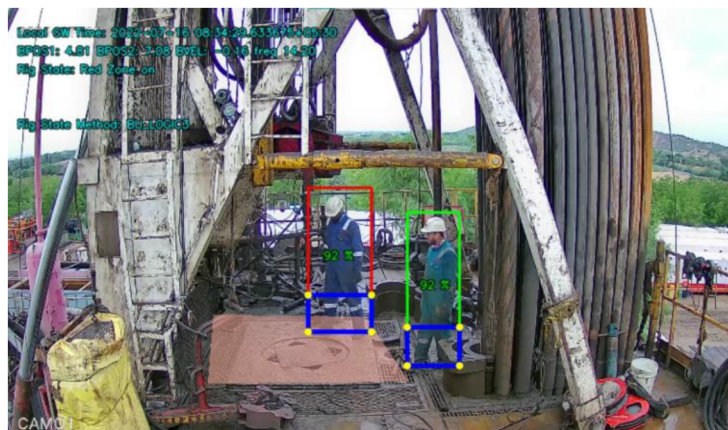
1. You will need to establish and assign the IP address of all the devices to ensure the Agora gateway, camera, Wi-Fi router and the shelly switch are interconnected.
2. Ensure all the IP addresses are assigned with the common server address – e.g. 192.168.1.10 (ensure the highlighted number of the above IP address is mentioned as "1" in all devices), if not you might need to login to the device and change to the same.
3. All the devices should be connected to the Agora gateway through the POE.
4. The Agora team will check the connection between them, and the connections are vital for proper functioning of the system
5. Ensure no loose or open connections
6. Ensure all the RJ 45 connectors are locked and are properly inserted

Note: At this stage you will need personnel with IT and camera connection experience/background. It is advisable that the IT personnel test the connection between the camera and the Agora systems in the office before installing it at the site. It is also advisable that the IT personnel stay at the site while the test is ongoing.

**Non-Compliance recording.** The following compliances are automatically detected from analyzing the video feed from the camera. Algorithms are developed to automatically detect the following non compliances and provide real time audio and visual alert on the rig floor and notification by emails. All the non-compliances are recorded in a table format with the picture of the non-compliance including the

date, time, type of violation and number of personnel making the non-compliances. And these violations and reports can be downloaded to discuss with the crew.

### ***TDS Related RED Zone area.***



**Figure 9—Analysis of the person and drilling parameter**

- Marked as 2m × 1.75m around the rotary hole (depending on the type of rig. Here it was defined based on the rig floor anti slippery mats and size of the TDS)
- Covers TDS area + Zone of Influence surrounding TDS area.

The video analytic solution detects any violation of the red zone area defined around TDS during operations (mainly drilling and tripping).

Any personnel breaching the TDS Red Zone area will be alerted and notified as critical non-compliance as per the operational conditions mentioned below. If any non-compliance occurs, i.e. if any person breaches the Red Zone, the event will be captured as a violation and a noncompliance alarm will be activated.

**Table 1—Drilling Parameters Logic**

| Drilling Parameters Logic to trigger the Red ZoneSr. No | Activity                               | RED ZONE TRIGGERING CRITERIA                                                     | RED ZONE EXCLUSION                                                     |
|---------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 1                                                       | Drilling / Circulation / Pressure test | Between 1 m to 29 m AND TDS Speed less than 0.15 m/s OR RPM > 0 OR SPPA > 25 psi | Slip In, Slip Out, Roughneck / Tongs engaging and disengaging Drilling |
| 2                                                       | Tripping/ Reaming/ Wash-down           | Between 1 m to 29 m Except - TDS Speed less than 0.15 m/s                        | Slip In, Slip Out, Roughneck / Tongs engaging and disengaging          |
| 3                                                       | Tripping (Single by Single)            | Greater than 1 m to 29 m Except - TDS Speed less than 0.15 m/s                   | Slip In, Slip Out, Roughneck / Tongs engaging and disengaging          |
| 4                                                       | Casing RIH/ Casing Wash-down           | Greater than 1 m to 29 m Except - TDS Speed less than 0.15 m/s                   | Slip In, Slip Out, Roughneck / Tongs engaging and disengaging          |

Note: The parameters may vary from Rig to Rig. The details need to be defined as Appendix 1 and need to be calculated at the rig specific thresholds for the logic, with the validation from the drilling operations team.

### ***Procedure to calculate the drilling parameters with example***

1. Set the minimum TDS block height and max TDS block height
2. Identify all tasks of each operation at the Rig floor like drilling, tripping in, tripping out, causing running in etc.

3. Identify the tasks when rig crew personnel are required to be inside the Red Zone for the respective operation
4. Observe/measure the time taken for each task considering the time taken by the TDS block height for placing slips, for torquing, etc.
5. Calculate the TDS Block speed for each operation
6. calculate the average minimum speed at which the TDS block moves for each operation
7. Repeat the above steps for each operation and calculate the average TDS speed
8. Fix/SET the minimum speed of the TDS block as the TDS Block Speed threshold

Note: Please refer to the table below as an example, the actual operations and the timings may differ as per activity.

**Table 2—Block height reference**

| TDS Block Height | Height in meter |
|------------------|-----------------|
| Max              | 1               |
| Min              | 30              |

Note: Based on the max height the TDS needs to travel, for example, since it is a triple stand rig, hence the fig of 30 m

### *Drilling Operations*

**Table 3—Examples of Parameters during Drilling operations**

| Operation         | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|-------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Open elevator     | 3.52                           | 102                           | 0                         | 0.00                                                      |
| Remove Hole cover | 3                              |                               | 4                         | -0.13                                                     |
| Placing Slips     | 2                              | 114                           | 11                        | -0.14                                                     |
| Doping            | 3                              | 31.5                          | 12                        | 1.00                                                      |
| Placing Tongs     | 15                             | 30.78                         | 15                        | 4.00                                                      |
| P/U DP            | 30.5                           | 32.8                          | 20                        | 3.10                                                      |
| Screwing the DP   | 30.5                           | 30.9                          | 22                        | 0.00                                                      |
| Remove Tongs      | 30.5                           | 30.9                          | 26                        | 0.00                                                      |
| Remove Slips      | 30.5                           | 103                           | 30                        | 0.00                                                      |

### *Casing operation*

**Table 4—Examples of Parameters during Casing operations**

| Operation          | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|--------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Placing Slips      | 5.5                            | 102                           | 1                         | 0.00                                                      |
| TDS Positioning    | 6.11                           | 31.08                         | 2                         | 0.00                                                      |
| Latching elevator  | 6.11                           | 31.8                          | 4                         | 0.00                                                      |
| Guiding of Casing  | 15.74                          | 31.55                         | 17                        | 0.74                                                      |
| Screwing of Casing | 15.73                          | 30.78                         | 22                        | 0.00                                                      |
| Tongs engagement   | 15.46                          | 30.47                         | 24                        | -0.14                                                     |
| Torquing Casing    | 15.46                          | 30.5                          | 26                        | 0.00                                                      |



| Operation        | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Removal of Tongs | 15.46                          | 95.12                         | 28                        | 0.00                                                      |
| Removal of Slips | 15.46                          | 100                           | 30                        | 0.00                                                      |

### *Tripping operation*

**Table 5—Examples of Parameters during Tripping operations**

| Operation                   | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|-----------------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Slips                       | 6.5                            | 30.5                          | 1                         | 0.00                                                      |
| Latching pipe from elevator | 31.05                          | 30.5                          | 9                         | 3.07                                                      |
| Applying Dope               | 31.05                          | 30.5                          | 10                        | 0.00                                                      |
| Guiding DP                  | 31.05                          | 30.5                          | 11                        | 0.00                                                      |
| Screwing DP                 | 31.05                          | 30.5                          | 12                        | 0.00                                                      |
| Engage Iron Roughneck       | 31.05                          | 32.03                         | 15                        | 0.00                                                      |
| Remove Slips                | 34.4                           | 67.3                          | 22                        | 0.48                                                      |

### *Tripping out*

**Table 6—Examples of Parameters during Tripping out operations**

| Operation         | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|-------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Elevator Latching | 4.5                            | 31.5                          | 0                         | 0.00                                                      |
| Slips Out         | 3.2                            | 20.7                          | 4                         | -0.13                                                     |
| Slips In          | 31.7                           | 32                            | 6                         | 14.25                                                     |
| Roughneck         | 31.59                          | 31.39                         | 8                         | -0.05                                                     |
| P/U DP            | 31.72                          | 32.5                          | 9                         | 0.13                                                      |
| Rack Stand        | 29.42                          | 31.39                         | 25                        | -0.14                                                     |

### *Tripping out Single by Single*

**Table 7—Examples of Parameters during tripping out singles by singles**

| Operation            | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|----------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Elevator Guiding     | 3.45                           | 30.78                         | 0                         | 0                                                         |
| elevator Latch       | 2.86                           | 31.45                         | 4                         | -0.15                                                     |
| Slips out            | 3.5                            | 31.45                         | 9                         | 0.13                                                      |
| P/U                  | 3.5                            | 31.45                         | 5                         | 0.00                                                      |
| Slips In             | 12.9                           | 31.39                         | 14                        | 1.04                                                      |
| R/N Engaged          | 12.62                          | 30.16                         | 12                        | 0.14                                                      |
| Guiding to Mousehole | 12.8                           | 31.7                          | 14                        | 0.09                                                      |



| Operation              | Block Height in m From Logging | Hook Load in Lbs From Logging | Time in seconds Stopwatch | TDS speed (Prev TDS Pos - Present TDS Pos) / (Time taken) |
|------------------------|--------------------------------|-------------------------------|---------------------------|-----------------------------------------------------------|
| Elevator Unlatch at MH | 2.75                           | 30.47                         | 31                        | -0.59                                                     |

## Results and Impact

Since implementation, the system has shown a significant reduction in unplanned red zone intrusions. Pilot testing across several land rigs demonstrated a 50–65% drop in weekly red zone violations within the first month of deployment. Crews reported heightened awareness and accountability, supported by the real-time feedback provided by the system.

Field validation also indicated improved supervisory capacity, as safety personnel could review archived footage to understand behavioral patterns and retrain specific crew members where necessary. This capability shifted the role of safety teams from reactive oversight to proactive coaching and planning.

By replacing subjective, inconsistent manual monitoring with standardized, data-driven detection, this system helps create a safer, more predictable work environment. It directly addresses the root cause of many rig-floor incidents and represents a scalable model for other high-risk work zones within energy operations.

Overall, video analytics for red zone detection serves as a cornerstone of digital PPE strategies. It demonstrates how AI can transform traditional safety protocols into intelligent, responsive systems capable of real-time intervention and long-term performance improvement.

## Platform for Automated Register Management

Manual lifting equipment register management in drilling and well services has long posed challenges in maintaining traceability, accuracy, and real-time compliance verification. Traditionally, the certification and inspection records for lifting tools such as slings, shackles, sheaves, spreader beams, and lifting frames are stored as hard-copy documentation or in spreadsheet-based trackers. These methods are prone to human error, redundancy, delayed updates, and fragmentation across multiple storage locations or devices.

When operating on fast-paced rigs, lifting equipment is often circulated or shared between crews and shifts. Without a unified and accessible verification system, crew members may unknowingly deploy equipment with expired certifications, missing documentation, or incompatible ratings—conditions that increase the likelihood of incidents during hoisting operations. Safety and operation teams may spend hours manually checking registers before commencing jobs, delaying operations and straining resources.

Further complicating this issue is the lack of standardization in certificate formats, with third-party vendors providing inspection reports in varied templates. This diversity makes manual extraction of key information—such as certificate numbers, expiration date, load rating, or tool description—laborious and inconsistent. In addition, the absence of unique equipment tracking using digital identifiers prevents the integration of historical usage logs or failure records, limiting the ability to perform proactive maintenance or retirement decisions based on data.

## Methodology

To address these challenges, a digital platform for automated register management was developed. The system leverages artificial intelligence and natural language processing to extract relevant metadata from PDF inspection certificates and automatically populate structured register fields. This not only standardizes the register structure but also ensures real-time accessibility, version control, and traceability.



Figure 10—Challenges with Manual extraction (old System)

The core of the platform is an intuitive digital interface accessible via web browsers or mobile applications. Users can upload inspection certificates to a centralized cloud repository, where optical character recognition (OCR) and rule-based AI models parse the document for key information. The system maps extracted values into a predefined register structure, associating each record with a unique serial number or QR code that physically tags the corresponding tool.

Users can search, sort, and filter the register by equipment category, tool type, location, or expiry status. Color-coded compliance indicators and automatic notifications alert supervisors when certificates are nearing expiry or missing. A built-in dashboard provides compliance metrics across rigs, contractors, and operational areas, enabling high-level monitoring by safety leadership.

### Technical Implementation

The system architecture integrates Microsoft Power Platform components, including Power Automate for workflow orchestration, SharePoint for secure data storage, Power BI for analytics, and PowerApps for the user interface. Data ingestion is automated via drag-and-drop or email uploads, with AI Builder or custom-trained models analyzing unstructured documents for consistency.

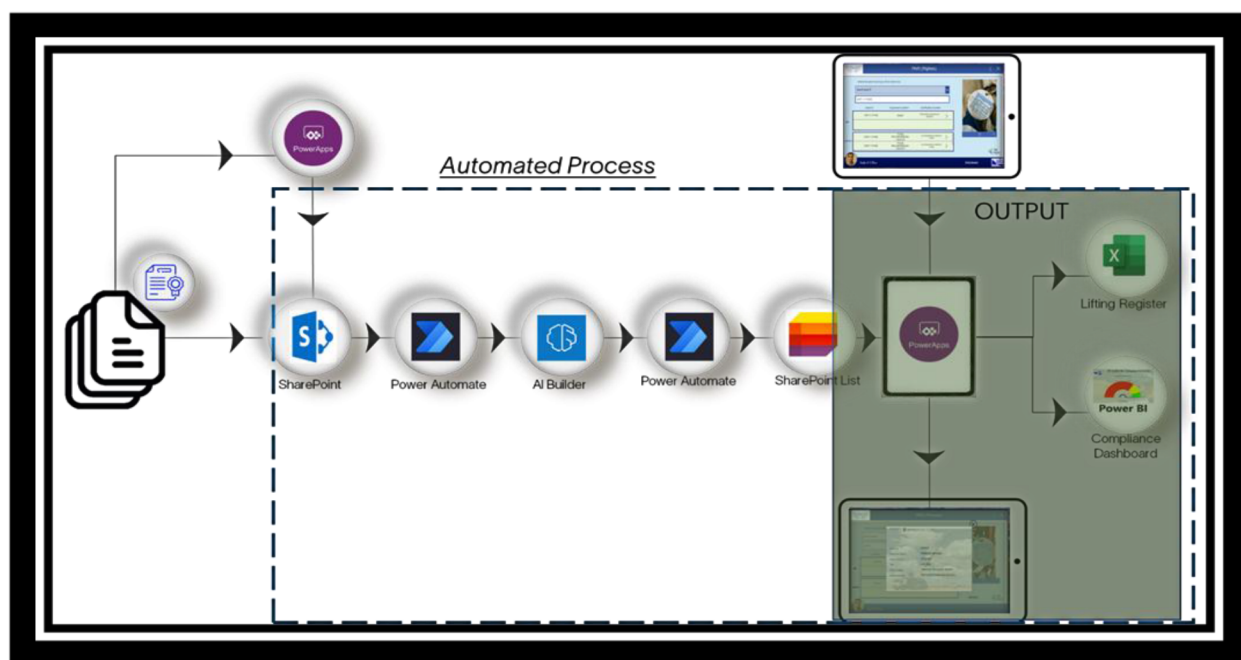


Figure 11—Process flow of PARI

Security is enforced through role-based access control, allowing different permission levels for site supervisors, HSE officers, and auditors. Activity logs track who accessed or modified each record, providing audit trails for compliance reviews. In offline environments, mobile users can still view cached records or scan Equipment tags for real-time validation, ensuring field functionality.

The digital platform is scalable, supporting multi-rig deployments with separate user domains or register sections while maintaining corporate-level visibility. Integration with enterprise asset management systems or enterprise resource planning (ERP) platforms.

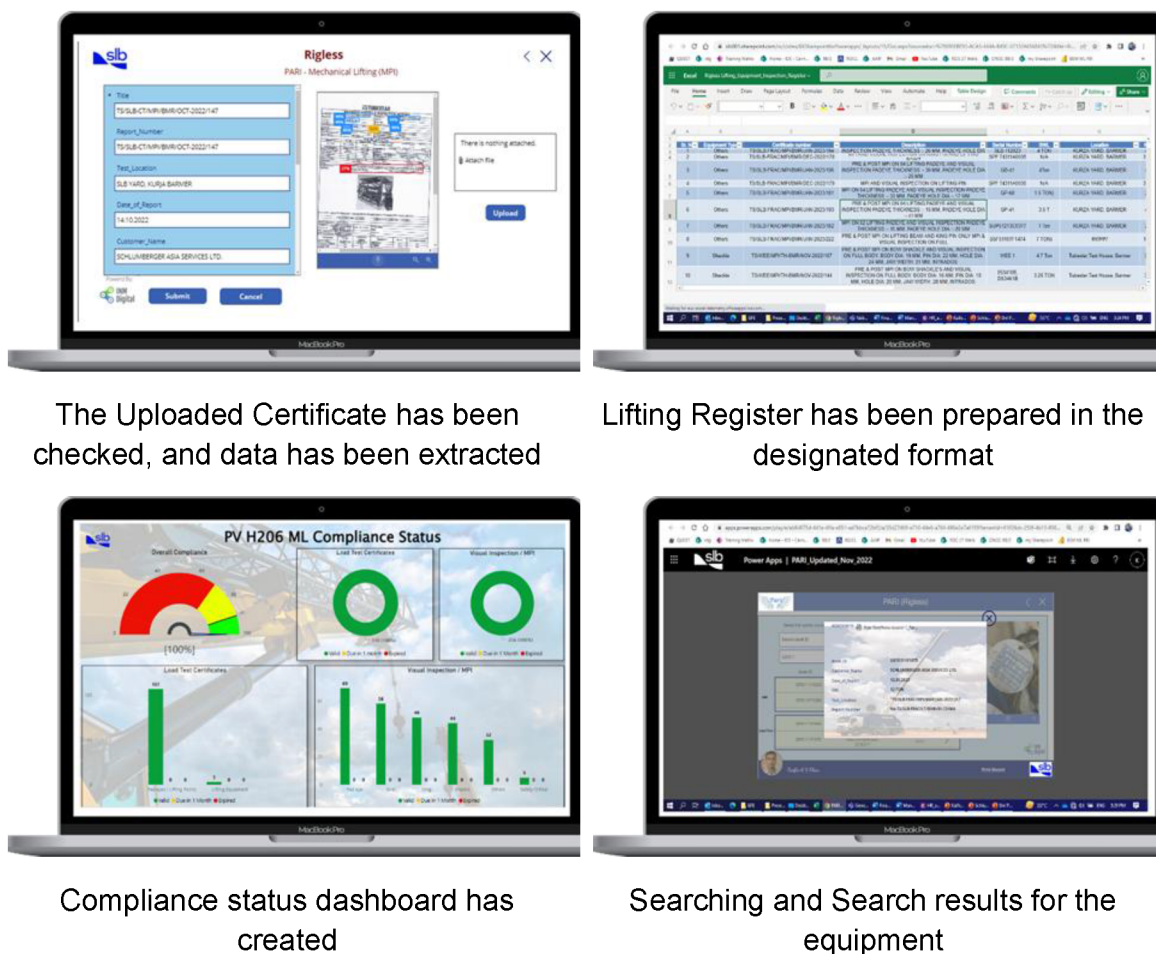


Figure 12—UI of PARI

## Results and Impact

Deployment of the automated register system has significantly improved lifting compliance and equipment traceability. Sites reported a 70% reduction in time required to validate tool readiness and a 90% reduction in errors associated with manual data transcription. Real-time access to digital registers enabled supervisors to make informed decisions on tool selection, replacement, or certification renewal before high-risk tasks.

Incident investigations also benefited from the availability of accurate, time-stamped records linking equipment use to operational context. This traceability enhanced root-cause analysis and informed preventative maintenance strategies. Training efforts improved, with digital registers used as visual aids to demonstrate proper lifting configurations and compliance expectations.

Moreover, the platform fostered a proactive compliance culture. Instead of reactive audits, safety teams could anticipate upcoming certification gaps and coordinate reinspection schedules accordingly.

The dashboard allowed leadership to benchmark contractors and identify systemic gaps, improving accountability across operations.

Automating the lifting register process through AI-driven document parsing and centralized digital access addresses a persistent operational risk. It streamlines compliance workflows, improves accuracy, and enhances readiness in lifting operations. As part of a broader digital PPE strategy, this platform reduces administrative load and empowers crews with the tools needed for safe, efficient lifting practices.

## Digital Observation Reporting Without Language Barriers

Safety observation reporting is a critical component of health, safety, and environment (HSE) systems, serving as an early warning mechanism to identify hazards and unsafe practices before they lead to incidents. However, effective implementation of observation systems in the oil and gas sector is often hindered by workforce diversity. In many rig operations, personnel speak a range of native languages, which introduces significant challenges in communication, particularly when reporting nuanced safety concerns.

Traditional observation cards or paper-based systems rely heavily on written narratives, which can be misinterpreted or remain unreported due to language constraints. This not only creates a barrier to participation but also introduces inaccuracies and delays in processing reported observations. Supervisors often need to manually translate submissions, leading to time-consuming administrative tasks and potential miscommunication. As a result, the safety system may become reactive rather than proactive, missing opportunities to intervene before incidents occur.

Additionally, paper-based forms are susceptible to physical damage, loss, and illegibility. In high-mobility job sites like drilling rigs, capturing, managing, and analyzing observation data using manual systems is neither scalable nor efficient. Without a consistent digital workflow, trends in safety performance cannot be easily visualized or communicated to leadership.

### Methodology

To overcome these limitations, a digital observation reporting platform was introduced. The system enables multilingual data entry through an intuitive mobile interface, automates language translation, and generates standardized safety reports. This ensures that all personnel—regardless of their language proficiency—can contribute to hazard identification and near-miss reporting.

The digital forms are accessible via QR codes or device-specific apps, enabling users to submit observations in their native language using voice-to-text or manual input. Submissions are instantly processed by integrated translation services, converting entries into a common language for safety officers to review. The translated content is stored in a structured digital format and categorized by type of observation, risk level, and operational location.

The platform can include options to attach photos, tag specific equipment or areas, and flag reports for immediate action. Supervisors receive alerts for high-priority observations and can assign follow-up tasks directly through the system. Dashboards visualize observation trends, enabling HSE teams to prioritize interventions and resource allocation.

### Technical Implementation

The platform was developed using Microsoft Forms for the front-end interface, Power Automate for backend workflows, and SharePoint for secure data storage. Automated translation is facilitated through integrated services that support multiple languages and dialects. The workflow triggers include data validation checks, routing of urgent reports to designated personnel, and real-time report generation.



Figure 13—UI of Auto Translated Multilingual Digital Observation Card

Each observation is assigned a unique identifier and timestamp, ensuring traceability and audit readiness. Data privacy is maintained through access controls and encryption protocols. Analytics are delivered through Power BI dashboards, which allow filtering by date, department, location, and severity.

For deployment in remote or offshore locations with limited connectivity, offline submission features enable local storage of observations until internet access is available. Once synced, the data integrates seamlessly into the central repository.

## Results and Impact

The adoption of the digital observation platform led to a significant increase in reporting participation rates, particularly from non-English-speaking crew members. Sites observed a 40% increase in total observations and a 55% rise in proactive reports (i.e., near-misses and unsafe conditions) within the first three months.

Translation automation reduced the average processing time per report by over 60%, freeing up safety officers to focus on data interpretation and corrective action rather than administrative translation tasks. Supervisors noted improved clarity in reported issues and faster resolution times due to the structured, digital nature of the data.

The availability of structured data also enhanced organizational learning. Trend analysis revealed recurring hazards and behavior-based safety concerns, allowing targeted coaching and job safety analyses (JSAs). Observational heatmaps highlighted operational zones requiring engineering or administrative controls.

In addition, the system contributed to building a culture of inclusivity and engagement. Personnel felt empowered to report without the fear of misunderstanding or language-related judgment. This inclusiveness supported a broader workforce morale and commitment to safety protocols.

Overall, the digital observation reporting solution represents a practical and scalable innovation. It strengthens frontline hazard detection, improves communication across diverse crews, and facilitates a responsive and data-informed safety management system. As a critical element of digital PPE strategy, this tool supports continuous improvement in workplace safety outcomes.



## Conclusion

The integration of artificial intelligence into personal protective systems represents a substantial advancement in occupational safety, especially in high-risk sectors such as oil and gas drilling. This paper presented three digital safety solutions that address key pain points within rig operations: red zone compliance, lifting equipment traceability, and inclusive hazard reporting. Each solution was designed to shift the safety paradigm from manual, reactive processes to automated, proactive systems.

The deployment of video analytics on the rig floor enabled real-time monitoring and alerting for red zone breaches, dramatically reducing the frequency of human error-driven violations. This system allowed for consistent enforcement of safety boundaries regardless of crew rotation, experience, or distraction levels. The result was a safer workspace characterized by increased awareness and standardized incident prevention practices.

The second innovation, the automated register management platform, tackled inefficiencies in lifting compliance. By transforming static documents into dynamic, searchable databases using AI-driven parsing, operational teams gained real-time visibility into certification status, tool availability, and inspection cycles. This not only reduced inspection times and administrative errors but also enhanced overall lifting safety by improving traceability.

The digital observation reporting system completed the safety framework by engaging a broader section of the workforce through multilingual input and automated translation. By facilitating greater participation and streamlining hazard communication, this tool bridged critical communication gaps and accelerated corrective actions. In doing so, it empowered frontline personnel and supported a culture of continuous improvement.

Collectively, these digital PPE systems enhance situational awareness, enforce compliance more effectively, and generate actionable insights for long-term safety improvements. They also reflect a scalable model that can be applied across varied geographic and operational contexts, making them valuable assets in enterprise HSE strategies.

As organizations aim to meet higher safety expectations, comply with global standards, and demonstrate ESG performance, such digital innovations will play an increasingly central role. The transition from manual to intelligent systems offers more than operational efficiency—it promises to reshape the future of occupational safety, enabling teams to predict, prevent, and perform at levels that were previously unachievable.

Future developments may include integration with wearable devices, voice recognition for real-time commands, or even predictive safety modeling based on machine learning. Continued collaboration between field teams, data scientists, and HSE professionals will be key to unlocking the full potential of these tools. In sum, digital PPE is not just a technological enhancement, it is a strategic evolution toward safer, smarter, and more resilient operations.

## Abbreviation

|            |                                                  |
|------------|--------------------------------------------------|
| AI         | Artificial Intelligence                          |
| ERP        | Enterprise Resource Planning                     |
| ESG        | Environmental Social Governance                  |
| FAR        | Fatal Accident Rate                              |
| HSE        | Health Safety and Environment                    |
| IOGP       | International Association of Oil & Gas Producers |
| IP address | Internet Protocol Address                        |
| IT         | Information Technology                           |
| JSA        | Job Safety Analysis                              |
| LWDC       | Lost Workdays Cases                              |

|          |                                                           |
|----------|-----------------------------------------------------------|
| m/s      | Meter per second                                          |
| OCR      | Optical Character Recognition                             |
| OPC      | OLE for Process Control (OLE: Object Linking & Embedding) |
| P/U DP   | Pick Up Drill Pipe                                        |
| Pari     | Platform for Automated Registers Intractability           |
| PLC      | Programmable Logic Controllers                            |
| POE      | Power Over Ethernet                                       |
| Power Bi | Power Business Intelligence                               |
| PSI      | Per Square Inch                                           |
| QR code  | Quick Response Code                                       |
| RIH      | Running in Hole                                           |
| RJ-45    | Registered Jack 45                                        |
| RPM      | Rotation Per Minute                                       |
| TDS      | Top Drive System                                          |
| UA       | Unified Architect                                         |
| UI       | User Interface                                            |
| Wi-Fi    | Wireless Fidelity                                         |

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